GY462 Real Estate Finance, Academic Year 2016/2017

Q1 – "The value of a real estate investment is an assessment of opportunities and risks."

a- Discuss this statement and explain how a discounted cash flow (DCF) method allows you to explicitly model risk and opportunities, and assess the value of a real estate asset. (5 points)

These issues were discussed during lecture 1 and touched upon in several other occasions.

b- Cap-rates are often used as short-cut methods to estimate the value of an investment. How are cap-rates applied to determine the *purchase* and *sale* price? Would you use the same cap-rate to determine *purchase* and *sale* value? Justify your answer and explain its rationale. (10 points)

These issues were the core of the first half of lecture 2.

c- In what sense can cap-rates as a valuation method suffer from myopia? What adjustments could you make to deal with this issue? (5 points)

As above: a significant amount of time was devoted to discussing this issue during lecture 2.

d- A fully-fledged DCF approach requires you to estimate all future income streams and costs. How can a pro-forma analysis assist with the task? What are the most important elements that you consider to estimate future Net Operating Incomes (NOI)? (15 points)

These issues were discussed at length during the second half of lecture 2.

a- Discuss elements of tenancy agreements and rent-roll structure that you can deploy strategically to maximise the value of your building by increasing the expected NOI. Enrich your answer by bringing in considerations presented by the 'case study' speakers. (20 points)

These issues were discussed at length during the second half of lecture 2 and then again during the external interventions in week 7 and week 10.

b- What are the main real-estate specific risks that make 'investing in buildings' directly different from investing in other financial assets? (10 points)

These points were covered at the beginning of lecture 1 and then again in lecture 3.

c- "Real Estate Investment Trusts (REITs) allow you to invest indirectly in real estate and benefit from its opportunities while bypassing it risks". Critically discuss this statement and explain the main characteristics of REITs – including constraints and common strategies for expansion and 'value adding'. (15 points)

REITS were the topic of lecture 6.

d- Discuss the experience and performance of UK REITs since their inception in 2007? Use the insights put forward during the intervention of the external lecturer. (20 points)

As above, these issues were discussed by both the lecturer and the external speaker during the lecture on REITS in week 7 (lecture 6).

Q2 – You need to borrow £500,000. You are offered by the bank the following mortgage: thirty year, fully amortizing Constant Payment Mortgage (CPM); monthly compounding; charging an annual interest of 8%.

a- What is the monthly payment owed to the bank of this mortgage? How much will have you paid in interest and amortization at the end of the first, second and third year? And what will be the outstanding balance on your mortgage at these dates? (10 points)

The monthly payment is £3668.82. The interest, amortization and outstanding balance at the end of years 1, 2 and 3 are as follows:

	Year 1	Year 2	Year 3
Interest	39,849	79,351.4	118,478.4
Amortization	4176.68	8700.3	13,599.26
Balance	495,823	491,299.7	486,400.7

b- Suppose the bank applies 5 discount points. What is the effective cost of borrowing if you hold to mortgage to maturity? And what is the effective cost of borrowing if you repay: *i*- after 10 years; *ii*- after 5 years; *iii*- after 3 years; and *iv*- after 1 year. (10 points)

The IRR at maturity is 8.5489%. In the other cases, students need to work out the outstanding balance at the different years. The figures are as follows:

Year	Outstanding balance	Implied IRR
1	495,823.179	13.39%
3	486400.74	9.958%
5	475,349.28	9.28%
10	438,623.5	8.788%

c- What do you notice from the calculations carried out at point -b? What is the bank's rationale for imposing discount points? And why would a borrower accept discount points? (5 points)

These issues were discussed at length during the second lecture on mortgages (lecture 5).

d- Consider now that the bank is applying quarterly compounding and quarterly payments to the same loan as in the heading to the question and 5 discount points. What would the effective cost of borrowing at maturity and at 10, 5, 3 and 1 year? And what if the bank was applying daily compounding (consider 365 days per year)? What do you notice? (10 points)

Calculations with quarterly compounding are as follows.

Constant payment: 11,024.05

Year	Outstanding balance	Implied IRR
1	495,779.27	13.44%
3	486,265.37	9.971%
5	475.118.32	9.287%
10	438,145.44	8.7913%

Calculations with daily compounding are as follows.

Constant payment: 120.53

Year	Outstanding balance	Implied IRR
1	495,844.55	13.36 %
3	486,466.66	9.95 %
5	475,461.79	9.28 %
10	438,856.69	8.79 %

I expect students to comment both on the patterns in the outstanding balance as well as the implied IRR when the frequency of payments changes. Issues with compounding were discussed in lecture 4 at the beginning of the mortgage topic.

e- Now consider again the original mortgage with monthly compounding. You agree with the bank to reduce payments to £3500 with the outstanding balance paid at maturity. The bank is still applying 5 discount points. What is the effective cost of borrowing of this loan if held to maturity? And what if you repay after 5 years? (10 points)

If holding to maturity, the "future value" of the loan with reduced payment is 251,606.76 (only partially amortizing). The implied IRR is 8.5% approx. If repaying after 5 years, the outstanding balance is 487,753.86 with an implied IRR of 9.266%.

f- You now agree to only amortize 60% of the loan, while leaving the other 40% to be repaid at maturity. What is the monthly payment of this mortgage? And if the bank still applies 5 discount points, what is the effective cost of borrowing at maturity and if you repay after 5 years? (10 points)

In this case, the future value of the loan at maturity remains 200,000 and the constant payment is 3534.6. This corresponds to an IRR at maturity with discount points of 8.5%. If repaying after 5 years, the outstanding balance is 485,209 with an implied IRR of 9.269%. These calculations give similar numbers to those found at the previous point. This is not surprising as the monthly payments are very close. I expect good students to notice and comment on this (even if not prompted).

g- You are now offered an alternative loan for the same amount of money (£500,000). The loan is still extending over 30 years with monthly compounding. However, this mortgage is a graduated payment mortgage (GPM) with a 5% increase in the monthly payment between the end of year 1 and beginning of year 2; the end of year 2 and beginning of year 3; and the end of year 3 and beginning of year 4. From then onwards, payments remain constant. What is the first monthly payment? And what are the monthly payments at the beginning of years 2, 3 and 4? What is the effective cost of borrowing of this loan if no discount points are charged? (20 points)

GPM were discussed during the second lecture on mortgages – lecture 5. The initial payment for this GPM can be thought of as coming from the following block DCF:

$$PV = PMT * \sum_{i=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{i}} + PMT\,(1+5\%) * \sum_{i=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{i}} * \frac{1}{\left(1+\frac{8\%}{12}\right)^{12}}$$

$$+ PMT\,(1+5\%)^2 * \sum_{i=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{i}} * \frac{1}{\left(1+\frac{8\%}{12}\right)^{24}} +$$

$$PMT\,(1+5\%)^3 * \sum_{i=1}^{324} \frac{1}{\left(1+\frac{8\%}{12}\right)^{i}} * \frac{1}{\left(1+\frac{8\%}{12}\right)^{36}} +$$

These expressions are all known except for PMT which is the unknown. The number can be found using the following identity:

$$500{,}000 = PMT * 11.496 + PMT * 1.05 * 11.496 * 0.9234 + PMT * 1.1025 * 11.496 * 0.8526 + PMT * 1.1576 * 132.577 * 0.7873 = PMT * 11.496 + PMT * 11.1462 + PMT * 10.8061 + PMT * 120.828 = PMT * 154.276$$

This gives an initial payment of 3240.94 then increasing to 3402.99, 3573.14 and finally 3751.798. If the student use these figures and the appropriate time windows to back out the IRR on this loan with no discount points, they should find 8% confirming that there results are correct.

h- Assume now the bank is charging 5 discount points. What is the effective cost of borrowing of this loan is held to maturity? And what is the loan is repaid at the end of year 3? And at the end of year 1? Provide intermediate steps for your calculations and discuss what you observe in terms of the principal reimbursement. (25 points)

If the loan is held to maturity, this is quite simple as the mortgage fully amortizes and this is just a DCF application. In this case the IRR is 8.5378%.

If the loan is pre-paid, students have to work out the outstanding balance. This cannot be done using the Amortization worksheet of the calculator. It needs to be done using the Future Value key in the time value of money worksheet. If the loan is repaid in one year, the balance increases to 501,150.29 as there is negative amortization. The implied IRR is 13.36%. The students then need to use 501,150.29 as the new balance for the loan paying 3402.99 monthly during the second year to find out the future value at the end of year 2. This is 500,378.54. This is now the new loan balance for year 3 during which the monthly payment is 3573.14. This gives a future value of 497,424. Using this as the outstanding balance at the end of year 3, the IRR is 9.933%.

This question is non-standard and meant to stretch the most able students. Approximate answers going in the right direction will be accepted as good for points.

Q3 – You are investing in the following real estate asset. The building has four floors and overall 20,000 square meters, equally divided across the four levels. The ground floor is currently used as office space but is not remunerative as it only passes £200 per square meter per year. The lease can be broken 6 months after purchase and you plan to do so. Following that, you expect to renovate the space for a period of one year. For the first 6 months you carry out structural work that cost £50 per square metre; for the following 6 months, you re-fit the space to accommodate a restaurant at the cost of £30 per square metre. At the beginning of the second year, you also pay £50,000 for architectural work and to obtain a change in planned use. After renovation is completed, you expect to rent the space to a restaurant at £300 per square metre per year. You estimate a vacancy rate of 50% for the first 6 months following completion of the works, but are confident the space will be fully let thereafter. As an incentive, you expect to give a 25% discount (concession) on the overall rent to the occupant for the first 18 months. However, from the beginning of year 4, you apply an overage charge of 10% of the revenues which are expected to be £3,000,000 per year. The first and second floors are used as office space and are remunerative. The occupant currently pays £250 per square metre per year, expected to increase to £280 from year 3. You envisage no changes for these two floors. Finally, the top floor is let to a consultancy currently paying £160 per square metre per year. This is below market level, but the lease is agreed for another 10 years with the rent adjusting to £180 in year 4. The consultancy also passes an overage income of 10% of its £1,500,000 yearly revenues. You are planning to hold the building for five years and, based on market conditions, you expect an exit yield of 10% at sale.

a- Using the information provided, present an annual discounted cash flow (DCF) model for this building (ignore within year discounting of costs and revenues). At what price are you expecting to sell the building? (20 points)

The operation DCF is as follows

		Year 1	Year 2	Year 3	Year 4	Year 5		Year 6
Floors 1&2	Rent	10000sqm * £250 = 2,500,000	10000sqm * £250 = 2,500,000	10000sqm * £280 = 2,800,000	10000sqm * £280 = 2,800,000	10000sqm * £280 = 2,800,000		10000sqm * £280 = 2,800,000
Total		*2,500,000*	*2,500,000*	*2,800,000*	*2,800,000*	*2,800,000*		*2,800,000*
Floor 3	Rent	5000sqm * £160 = 800,000	5000sqm * £160 = 800,000	5000sqm * £160 = 800,000	5000sqm * £180 = 900,000	5000sqm * £180 = 900,000		5000sqm * £180 = 900,000
	Overage charge	10% * 1,500,000 = 150,000	10% * 1,500,000 = 150,000	10% * 1,500,000 = 150,000	10% * 1,500,000 = 150,000	10% * 1,500,000 = 150,000		10% * 1,500,000 = 150,000

Total		*950,000*	*950,000*	*950,000*	*1,050,000*	*1,050,000*		*1,050,000*
Ground floor	Rent	5000sqm * £200 * 0.5 years = 500,000	5000sqm * £300 * 0.5 years * 50% vacancy prob. = 375,000	5000 * £300 = 1,500,000	5000 * £300 = 1,500,000	5000 * £300 = 1,500,000		5000 * £300 = 1,500,000
	Overage charge	0	0	0	10% * £3,000,000 = 300,000	10% * £3,000,000 = 300,000		10% * £3,000,000 = 300,000
	Concessions	0	-93750	-375,000	0	0		0
	Work costs	5000sqm * £50 = -250,000	5000sqm * £30 + 50,000 = -200,000	0	0	0		0
Total		*250,000*	*81,250*	*1,125,000*	*1,800,000*	*1,800,000*		*1,800,000*
Grand Total		***3,700,000***	***3,531,250***	***4,875,000***	***5,650,000***	***5,650,000***		***5,650,000***

The sale value is £56,500,000.

b- If you are asked to pay £37,000,000 for the building, what is the implied Internal Rate of Return (IRR) of the investment? (5 points)

The IRR is 19%.

c- Analysts in practice often "compress" the exit yield. What does it mean? Why is this done and what are the drawbacks of doing this? Discuss this issue in general terms and then specifically comment on the IRR you obtain if you apply an exit yield of 8% and an exit yield of 12% to your cash flow analysis. (15 points)

The first case study speaker discussed yield compression. This is a way to generate extra returns although it brings risks and so this assumption is often flexed (i.e., scenarios are created and considered). The IRR with 8% yield is 23.4% (sale price 70,625,000) while using a 12% yield is 16.1% (sale price 47,083,333). I expect the students to discuss how the IRR moves depending on the more/less conservative assumptions about the exit yield.

d- Assume you now borrow to fund your investment. You obtain an interest only loan with an LTV of 50% and charging a 10% annual interest. What is the IRR of the leveraged investment using a fully-fledged DCF approach (using a 10% exit yield)? Do you obtain the

same figure if you apply a Weighted Average Costs of Capital (WACC) short-cut approach? Comment on your findings. (10 points)

The loan covers 18,500,000 of the investment and attracts a 1,850,000 annual interest charge. The IRR with the loan is 26.3% - an increase relative to the ungeared IRR. The WACC answer would be 19% + (19% - 10%) * 1 (=D/E) = 28%. The discrepancy is due to changes in the value of the equity over time.

e- Flex again your exit yield to be 8% and 12%. What happens to the IRR of the leveraged investment in these two cases? Comment on your findings. (5 points)

In this case, the IRRs become 32.63% and 21%. Returns are still amplified, but the difference (variability) in the conservative/aggressive yield scenarios widens. This highlights the role of leverage in amplifying risks.

f- Now imagine that you work for a speculative fund which shortens your holding period to three years. While the same assumptions hold for the ground, first and second floors, the fund decides to break the lease with the top floor tenant and clear the space in the first year after purchase. Litigations with the tenant last a whole year and cost £1,500,000 in legal fees. During this period, no income or overage charge is paid by the tenant. Once the tenant has left (at the end of year one), you renovate the floor for a whole year at the cost of £50 per square metre. Following that, however, you expect to lease to the space to another consultancy for £300 per square metre per year for ten years. In order to sign this deal, you offer 18 months free of rent to the company. You expect your exit yield to be 8% and use the same lending facility described above. You are asked to pay the same price. What is your IRR? (20 points)

In this case, the operation DCF is as follows (note, floors 1&2 and ground floor are the same)

		Year 1	Year 2	Year 3		Year 4
Floors 1&2, Total		*2,500,000*	*2,500,000*	*2,800,000*		*2,800,000*
Floor 3	Rent	0	0	0		5000sqm * £300 * 0.5 (6 months rent free) = 750,000
	Litigation	-1,500,000	0	0		0
	Refurbishment costs	0	5000sqm * £50 = -250,000	0		0
Total		*-1,500,000*	*-250,000*	*0*		*750,000*

Ground floor, Total		*250,000*	*81,250*	*1,125,000*		*1,800,000*
Grand Total		***1,250,000***	***2,331,250***	***3,925,000***		***5,350,000***

The sale value is £66,875,000. This gives an IRR of 39.25 %

g- Imagine the economy significantly worsens. In this case, prevailing exit yields are 12%. Moreover, the deal for the renovated top floor falls through and the space remains vacant in year three and four. What happens to your leveraged IRR? (15 points)

In this case, the NOI of the third floor in Year 4 is zero so the overall Year 4 NOI is 4,600,000. With an exit yield of 12% this gives a sale price of 38,333,333 and an overall IRR of 5.54 %.

h- Consider again the two scenarios at points f- and g-. What are the respective IRRs if you do not use leverage? Comment on your findings highlighting the role of leverage in magnifying risks and returns. (10 points)

Considering the two cases above without leverage gives IRRs of respectively 26.99 % and 7.70 %. Clearly this shows that the returns are magnified by leveraging – but also that the margin of variation is much wider in the "good" and "bad" case. Stated differently, risk has been amplified too.